Assignment: AI-Based Analysis of Drone Imagery

Objective

Develop a Python-based pipeline that analyses drone-acquired imagery to detect and classify land surface features using AI or deep learning techniques.

Note: *The task simulates a real-world workflow where drone imagery is used for environmental monitoring, agriculture, or infrastructure inspection.*

Problem Statement

- You are provided with high-resolution drone imagery of an urban environment.
- Your task is to build a Python application that:
  1. Loads and processes the drone imagery.
  2. Extracts useful image features.
  3. Uses AI/DL techniques to classify regions into different land-cover categories.
  4. Visualizes and exports the results.

Dataset Link :

https://mega.nz/file/QWQCBKSa#641HnSSoS_g8fUvzncdKqCqNkEYczMaolaDW27SzrQY

Required Tasks

1. Data Loading & Preprocessing

Write Python code to:

* Load drone imagery (GeoTIFF / JPEG / PNG).

* Handle large image sizes efficiently.

* Perform preprocessing such as:

    * Resizing / Tiling

    * Normalization

    * Noise removal.

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Suggested libraries:

* Numpy
* Opencv
* Rasterio
* Pillow

2. Feature Extraction

Extract meaningful features from the imagery.

Examples:

* Texture features (GLCM / entropy)

* Vegetation indices (if NIR available)

* Edge features

3. AI / Machine Learning Model

Train a model to classify image regions.

Possible approaches:

Artificial Intelligence

* Segment Anything Model

* Detectron2

Deep Learning (Optional)

* CNN classifier

* Transfer learning using pretrained models

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4. Image Classification

Divide the drone image into patches or tiles and classify each region.

Example classes:

* Vegetation

* Soil

* Water

* Built structures

* Roads

Output should be a classification map.

5. Visualization

Visualize results using Python.

Example:

* Classification overlay

* color-coded maps

* Prediction confidence maps

6. Export Results

Export outputs in one or more formats:

* PNG visualization

* CSV prediction table

* GeoTIFF classification map (optional)

Expected Project Structure

```
drone_ai_assignment/
data/
drone_image.tif
src/
  data_loader.py
  preprocessing.py
  feature_extraction.py
  model_training.py
  prediction.py
  visualization.py
```

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main.py

requirements.txt

README.md

Deliverables

Candidates must submit:

1. Python source code

2. requirements.txt

3. README with:

  * Installation instructions

  * Explanation of approach

Note: The use of AI tools is restricted to a maximum of 30%. Any usage beyond this limit will result in a negative impact on the evaluation score.

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